

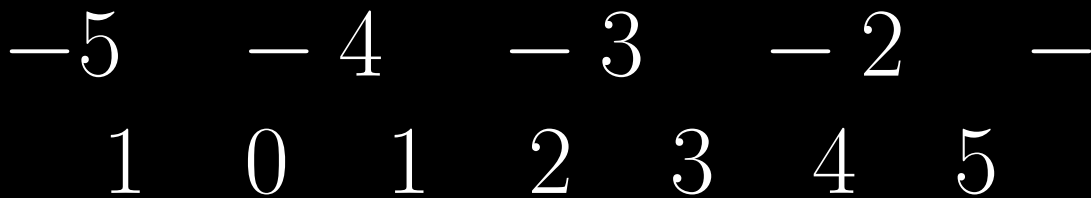
Comparing and Ordering Integers

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Integers include the **negative integers**, **zero**,
and **positive integers**.

Using a Number Line

A **number line** helps compare the values of integers.



Right = Greater

An integer located farther to the **right** on the number line is **greater**.

0 1 2 3 $\longrightarrow$ *greater*

Left = Less

An integer located farther to the **left** on the number line is **less**.

less $\longleftarrow$ -3 - 2 - 1 0

Comparison Symbols

Use the symbols $>$, $<$, and $=$ to compare integers.

Symbol	Meaning
$>$	greater than
$<$	less than
$=$	equal to

Ordering Integers

To **order integers**, arrange them from:

- ▶ **least to greatest**, or
- ▶ **greatest to least**

Tricky Negatives

Negative numbers with larger absolute values are actually **smaller**.

$$-9 < -4$$

Even though $9 > 4$, on the number line -9 is farther left, so it is smaller.

Examples

Example 1

Compare: -6 _____ 2

$$-6 < 2$$

-6 is to the left of 2 on the number line.

Example 2

Compare: -3 ____ -10

$$-3 > -10$$

Even though $10 > 3$, -10 is farther left, so it is smaller.

Example 3

Arrange from least to greatest: 6, -3 , 0, -8

-8 , -3 , 0, 6

Example 4

Arrange from greatest to least: $-5, 9, -1, 4$

$9, 4, -1, -5$

Practice Exercises

Practice 1

Compare: -7 3

Practice 2

Compare: 5 -2

Practice 3

Compare: -8 -5

Practice 4

Compare: 0 -6

Practice 5

Compare: 4 4

Practice 6

Arrange from least to greatest: -2 , 5 , -7 , 0

Practice 7

Arrange from greatest to least: 8, -1 , 3, -6

Practice 8

Arrange from least to greatest: -10 , -3 , 2 , 7

Practice 9

Arrange from greatest to least: 0, -9 , 6, -2

Practice 10

Arrange from least to greatest:

9, -4 , 1, -8 , 0

Activities

Activity 1

Compare: -9 2

Activity 2

Compare: $6 \underline{\hspace{1cm}} -5$

Activity 3

Compare: -3 -11

Activity 4

Compare: 0 _____ 7

Activity 5

Compare: -1 -1

Activity 6

Arrange from least to greatest: 4, -6 , 1, -3

Activity 7

Arrange from greatest to least: -4 , 9 , 2 , -8

Activity 8

Arrange from least to greatest: 5, -9 , 0, -2

Activity 9

Arrange from greatest to least: 7, -5 , 3, 0

Activity 10

Arrange from least to greatest:

$-12, 8, -1, 4, 0$

Thank you!